

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Trigonometric Square Root Inverse

Evaluate the definite integral:

$$\int_0^{\pi/4} \sec(x) \sqrt{\sec(2x)} dx$$

Solution: Let the integral be I . We first rewrite the integrand using cosines:

$$I = \int_0^{\pi/4} \frac{1}{\cos(x)} \frac{1}{\sqrt{\cos(2x)}} dx$$

Using the double-angle identity $\cos(2x) = 1 - 2\sin^2(x)$, substitute this into the denominator and multiply the numerator and denominator by $\cos(x)$:

$$I = \int_0^{\pi/4} \frac{\cos(x)}{\cos^2(x) \sqrt{1 - 2\sin^2(x)}} dx$$

Since $\cos^2(x) = 1 - \sin^2(x)$, the integral becomes:

$$I = \int_0^{\pi/4} \frac{\cos(x)}{(1 - \sin^2(x)) \sqrt{1 - 2\sin^2(x)}} dx$$

Apply the substitution $u = \sqrt{2}\sin(x)$, which gives $du = \sqrt{2}\cos(x) dx$, or $\cos(x) dx = \frac{du}{\sqrt{2}}$. The bounds change as follows: when $x = 0$, $u = 0$; when $x = \pi/4$, $u = \sqrt{2}\sin(\pi/4) = 1$. Substitute u and du into the integral:

$$I = \int_0^1 \frac{1}{\left(1 - \frac{u^2}{2}\right) \sqrt{1 - u^2}} \frac{du}{\sqrt{2}} = \sqrt{2} \int_0^1 \frac{1}{(2 - u^2) \sqrt{1 - u^2}} du$$

Now, apply a trigonometric substitution by letting $u = \sin(\theta)$. Then $du = \cos(\theta) d\theta$. The bounds change from $[0, 1]$ to $[0, \pi/2]$.

$$I = \sqrt{2} \int_0^{\pi/2} \frac{\cos(\theta)}{(2 - \sin^2(\theta)) \sqrt{1 - \sin^2(\theta)}} d\theta$$

Since $\sqrt{1 - \sin^2(\theta)} = \cos(\theta)$ for $\theta \in [0, \pi/2]$, the cosines cancel out perfectly:

$$I = \sqrt{2} \int_0^{\pi/2} \frac{1}{2 - \sin^2(\theta)} d\theta$$

To evaluate this, divide the numerator and the denominator by $\cos^2(\theta)$:

$$I = \sqrt{2} \int_0^{\pi/2} \frac{\sec^2(\theta)}{2\sec^2(\theta) - \tan^2(\theta)} d\theta$$

Use the identity $\sec^2(\theta) = 1 + \tan^2(\theta)$ in the denominator:

$$I = \sqrt{2} \int_0^{\pi/2} \frac{\sec^2(\theta)}{2(1 + \tan^2(\theta)) - \tan^2(\theta)} d\theta = \sqrt{2} \int_0^{\pi/2} \frac{\sec^2(\theta)}{2 + \tan^2(\theta)} d\theta$$

Finally, let $t = \tan(\theta)$, so $dt = \sec^2(\theta) d\theta$. The limits change to $[0, \infty)$:

$$I = \sqrt{2} \int_0^{\infty} \frac{1}{2 + t^2} dt = \sqrt{2} \left[\frac{1}{\sqrt{2}} \arctan \left(\frac{t}{\sqrt{2}} \right) \right]_0^{\infty}$$

Evaluating at the limits:

$$I = 1 \cdot \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{2}$$

1.2 Hard Integral

Exponential Gaussian Symmetries

Evaluate the integral:

$$\int_0^{\infty} \left[e^{-x^2-5x} + e^{-x^2+5x} \right] dx$$

Solution: Let the given integral be I . We can split it into two separate integrals:

$$I = \int_0^{\infty} e^{-x^2-5x} dx + \int_0^{\infty} e^{-x^2+5x} dx$$

Consider the second integral. Apply the substitution $u = -x$, so $du = -dx$. The limits of integration change from $[0, \infty)$ to $[0, -\infty)$:

$$\int_0^{\infty} e^{-x^2+5x} dx = \int_0^{-\infty} e^{-(-u)^2+5(-u)} (-du) = \int_{-\infty}^0 e^{-u^2-5u} du$$

Since u is a dummy variable, we can replace it with x :

$$\int_0^{\infty} e^{-x^2+5x} dx = \int_{-\infty}^0 e^{-x^2-5x} dx$$

Now, substitute this back into our expression for I :

$$I = \int_0^{\infty} e^{-x^2-5x} dx + \int_{-\infty}^0 e^{-x^2-5x} dx = \int_{-\infty}^{\infty} e^{-x^2-5x} dx$$

To evaluate this integral over the entire real line, complete the square in the exponent:

$$-x^2 - 5x = - \left(x^2 + 5x + \frac{25}{4} \right) + \frac{25}{4} = - \left(x + \frac{5}{2} \right)^2 + \frac{25}{4}$$

Substitute this back into the integral:

$$I = \int_{-\infty}^{\infty} e^{-(x+5/2)^2+25/4} dx = e^{25/4} \int_{-\infty}^{\infty} e^{-(x+5/2)^2} dx$$

Let $w = x + 5/2$, so $dw = dx$. The limits remain unchanged:

$$I = e^{25/4} \int_{-\infty}^{\infty} e^{-w^2} dw$$

This is the standard Gaussian integral, which evaluates to $\sqrt{\pi}$. Thus:

$$I = e^{25/4} \sqrt{\pi}$$

Differentials

2.1 Medium Differential

First-Order ODE with Initial Condition

Solve the following initial value problem:

$$y' = 2t^{-1} + e^{-y}$$

Given $y(1) = 0$, find the exact value of $y(1.5)$.

Solution: Multiply the entire differential equation by e^y to clear the exponential from the right side:

$$e^y y' = \frac{2}{t} e^y + 1$$

Apply the substitution $u = e^y$. Differentiating with respect to t gives $u' = e^y y'$. Substitute this into the equation:

$$u' = \frac{2}{t} u + 1 \implies u' - \frac{2}{t} u = 1$$

This is a standard first-order linear differential equation. We find the integrating factor $\mu(t)$:

$$\mu(t) = \exp\left(\int -\frac{2}{t} dt\right) = \exp(-2 \ln t) = t^{-2}$$

Multiply the linear ODE by t^{-2} :

$$t^{-2} u' - 2t^{-3} u = t^{-2}$$

The left side condenses by the product rule:

$$\frac{d}{dt} (t^{-2} u) = t^{-2}$$

Integrate both sides with respect to t :

$$t^{-2} u = -t^{-1} + C \implies u = -t + Ct^2$$

Since $u = e^y$, we substitute back:

$$e^y = -t + Ct^2$$

Use the initial condition $y(1) = 0$ to solve for C :

$$e^0 = -(1) + C(1)^2 \implies 1 = -1 + C \implies C = 2$$

The specific solution to the differential equation is:

$$e^y = 2t^2 - t \implies y(t) = \ln(2t^2 - t)$$

To find $y(1.5)$, substitute $t = 1.5 = \frac{3}{2}$:

$$y(1.5) = \ln\left(2\left(\frac{3}{2}\right)^2 - \frac{3}{2}\right) = \ln\left(2\left(\frac{9}{4}\right) - \frac{3}{2}\right) = \ln\left(\frac{9}{2} - \frac{3}{2}\right) = \ln\left(\frac{6}{2}\right) = \ln(3)$$

2.2 Hard Differential

Orthogonal Trajectory Family

Let φ be a family of curves orthogonal to a family of trajectories ζ which satisfies the following equation:

$$e^y \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) = Ce^{\pi e}$$

Given that $\varphi(0) = 0$, find the exact value of $\pi\varphi\left(\frac{\pi}{2}\right)$. Assume continuity over the interval.

Solution: First, we find the differential equation representing the family of trajectories ζ . Differentiate the given equation implicitly with respect to x :

$$\frac{dy}{dx} e^y \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) + e^y \cdot \frac{1}{2} \sec^2\left(\frac{x}{2} + \frac{\pi}{4}\right) = 0$$

Divide through by e^y (since $e^y \neq 0$):

$$y' \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) + \frac{1}{2} \sec^2\left(\frac{x}{2} + \frac{\pi}{4}\right) = 0$$

Solve for y' to find the slope of ζ :

$$y' = -\frac{\sec^2\left(\frac{x}{2} + \frac{\pi}{4}\right)}{2 \tan\left(\frac{x}{2} + \frac{\pi}{4}\right)} = -\frac{1}{2 \sin\left(\frac{x}{2} + \frac{\pi}{4}\right) \cos\left(\frac{x}{2} + \frac{\pi}{4}\right)}$$

Using the double angle identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ where $\theta = \frac{x}{2} + \frac{\pi}{4}$:

$$y' = -\frac{1}{\sin\left(x + \frac{\pi}{2}\right)}$$

Since $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$, the differential equation for ζ simplifies to:

$$\frac{dy}{dx} = -\frac{1}{\cos(x)} = -\sec(x)$$

The family of orthogonal trajectories φ must have a slope that is the negative reciprocal of the slope of ζ . Let the slope of φ be y'_{ort} :

$$y'_{ort} = -\frac{1}{-\sec(x)} = \cos(x)$$

Integrate this to find the equation for the curves φ :

$$\varphi(x) = \int \cos(x) dx = \sin(x) + K$$

Use the initial condition $\varphi(0) = 0$ to find the constant K :

$$\sin(0) + K = 0 \implies K = 0$$

Thus, the specific orthogonal trajectory is exactly $\varphi(x) = \sin(x)$. We must find the value of $\pi\varphi(\pi/2)$:

$$\pi\varphi\left(\frac{\pi}{2}\right) = \pi \sin\left(\frac{\pi}{2}\right) = \pi(1) = \pi$$

Derivatives

3.1 Medium Derivative

Rocket Sled Minimum Acceleration

A rocket sled position in metres after x seconds is:

$$s(x) = -(x-6)(x-2)(x^3 - 2x^2 - 4x + 4)$$

Determine the minimum acceleration of the sled.

Solution: First, expand the polynomial representing the position $s(x)$. Note that $(x-6)(x-2) = x^2 - 8x + 12$. Multiplying this with the cubic polynomial:

$$s(x) = -(x^2 - 8x + 12)(x^3 - 2x^2 - 4x + 4)$$

$$s(x) = -[x^2(x^3 - 2x^2 - 4x + 4) - 8x(x^3 - 2x^2 - 4x + 4) + 12(x^3 - 2x^2 - 4x + 4)]$$

$$s(x) = -[(x^5 - 2x^4 - 4x^3 + 4x^2) + (-8x^4 + 16x^3 + 32x^2 - 32x) + (12x^3 - 24x^2 - 48x + 48)]$$

Combine like terms to yield the fully expanded position function:

$$s(x) = -x^5 + 10x^4 - 24x^3 - 12x^2 + 80x - 48$$

Velocity $v(x)$ is the first derivative of $s(x)$:

$$v(x) = s'(x) = -5x^4 + 40x^3 - 72x^2 - 24x + 80$$

Acceleration $a(x)$ is the second derivative of $s(x)$:

$$a(x) = v'(x) = -20x^3 + 120x^2 - 144x - 24$$

To find the minimum acceleration, we find the critical points by taking the derivative of $a(x)$ and setting it to zero:

$$a'(x) = -60x^2 + 240x - 144 = 0$$

Divide the entire equation by -12 :

$$5x^2 - 20x + 12 = 0$$

Using the quadratic formula to solve for x :

$$x = \frac{20 \pm \sqrt{(-20)^2 - 4(5)(12)}}{2(5)} = \frac{20 \pm \sqrt{400 - 240}}{10} = \frac{20 \pm \sqrt{160}}{10} = 2 \pm \frac{2\sqrt{10}}{5}$$

Because the leading coefficient of $a(x)$ is negative ($-20x^3$), the cubic function goes from $+\infty$ down to a local minimum, up to a local maximum, and down to $-\infty$. The local minimum corresponds to the smaller root:

$$x_{min} = 2 - \frac{2\sqrt{10}}{5}$$

To evaluate $a(x_{min})$ easily, we can divide the polynomial $a(x)$ by $5x^2 - 20x + 12$. The remainder of this division provides a linear function for the values at the roots:

$$a(x) = (-4x + 8)(5x^2 - 20x + 12) + (64x - 120)$$

At $x = x_{min}$, the quadratic factor is zero, leaving only the remainder:

$$a(x_{min}) = 64 \left(2 - \frac{2\sqrt{10}}{5} \right) - 120 = 128 - \frac{128\sqrt{10}}{5} - 120 = 8 - \frac{128\sqrt{10}}{5}$$

The minimum acceleration is exactly $\frac{40-128\sqrt{10}}{5} \text{ m/s}^2$.

3.2 Hard Derivative

Alternating Taylor Series Sum

Let $f(x) = \sqrt{1 + \cos(x)}$. Compute $\sum_{n=0}^{\infty} f^{(n)}\left(\frac{\pi}{2}\right)$.

Solution: Using the half-angle identity $1 + \cos(x) = 2 \cos^2\left(\frac{x}{2}\right)$, the function can be rewritten for $x \in [0, \pi]$ as:

$$f(x) = \sqrt{2} \cos\left(\frac{x}{2}\right)$$

We compute the first few derivatives of $f(x)$:

$$f(x) = \sqrt{2} \cos\left(\frac{x}{2}\right)$$

$$f'(x) = -\frac{\sqrt{2}}{2} \sin\left(\frac{x}{2}\right)$$

$$f''(x) = -\frac{\sqrt{2}}{4} \cos\left(\frac{x}{2}\right)$$

$$f'''(x) = \frac{\sqrt{2}}{8} \sin\left(\frac{x}{2}\right)$$

$$f^{(4)}(x) = \frac{\sqrt{2}}{16} \cos\left(\frac{x}{2}\right)$$

Evaluate these derivatives at $x = \frac{\pi}{2}$:

$$f\left(\frac{\pi}{2}\right) = \sqrt{2} \cos\left(\frac{\pi}{4}\right) = \sqrt{2} \left(\frac{1}{\sqrt{2}}\right) = 1$$

$$f'\left(\frac{\pi}{2}\right) = -\frac{\sqrt{2}}{2} \sin\left(\frac{\pi}{4}\right) = -\frac{1}{2}$$

$$f''\left(\frac{\pi}{2}\right) = -\frac{\sqrt{2}}{4} \cos\left(\frac{\pi}{4}\right) = -\frac{1}{4}$$

$$f''' \left(\frac{\pi}{2} \right) = \frac{\sqrt{2}}{8} \sin \left(\frac{\pi}{4} \right) = \frac{1}{8}$$

This generates a predictable sequence. The values of $f^{(n)}(\pi/2)$ for $n \geq 0$ are: $1, -\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, -\frac{1}{32}, -\frac{1}{64}, \dots$. The signs follow the repeating pattern $\{+, -, -, +, \dots\}$, which corresponds to the real and imaginary components of successive powers of i . We can write the sum as the real part of an infinite geometric series in the complex plane. Let the sum be $S = \sum_{n=0}^{\infty} a_n$. Notice that we can pair terms $2k$ and $2k+1$:

$$a_{2k} + a_{2k+1} = \frac{(-1)^k}{4^k} + \frac{(-1)^{k+1}}{2 \cdot 4^k} = \frac{(-1)^k}{4^k} \left(1 - \frac{1}{2} \right) = \frac{1}{2} \left(-\frac{1}{4} \right)^k$$

Thus, the sum S is equivalent to a geometric series with first term $a = \frac{1}{2}$ and common ratio $r = -\frac{1}{4}$:

$$S = \sum_{k=0}^{\infty} \frac{1}{2} \left(-\frac{1}{4} \right)^k = \frac{1/2}{1 - (-1/4)} = \frac{1/2}{5/4} = \frac{2}{5}$$

Physics & Engineering

4.1 Mechanics

Kinematics of a Cart

At time $t = 0$, a cart is at position $x = 10$ m and has a velocity of 3 m s^{-1} in the $-x$ -direction. The cart has a constant acceleration in the $+x$ -direction with magnitude $3 < a < 6 \text{ m s}^{-2}$. What is the range of the position of the cart at $t = 1$ s?

Solution: The cart undergoes motion with constant acceleration, which can be described by the kinematic equation for position:

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

From the problem description, we are given the following initial conditions: Initial position, $x_0 = 10$ m Initial velocity, $v_0 = -3 \text{ m s}^{-1}$ (negative because it travels in the $-x$ -direction) Time, $t = 1$ s Substitute these into the kinematic equation to find the position at $t = 1$:

$$x(1) = 10 + (-3)(1) + \frac{1}{2} a (1)^2 = 7 + 0.5a$$

We are given an inequality for the constant acceleration: $3 < a < 6$. Substitute these limits into the equation for $x(1)$ to find the minimum and maximum possible positions. For the lower bound of acceleration ($a = 3$):

$$x_{\min} = 7 + 0.5(3) = 7 + 1.5 = 8.5 \text{ m}$$

For the upper bound of acceleration ($a = 6$):

$$x_{\max} = 7 + 0.5(6) = 7 + 3 = 10 \text{ m}$$

Therefore, the range of the position of the cart at $t = 1$ s is between 8.5 m and 10 m. This can be expressed as $(8.5, 10)$ m or $8.5 \text{ m} < x < 10 \text{ m}$.

4.2 Quantum Mechanics

U-235 Gamma Ray Count Rate

A cylindrical uranium pellet contains 4.00 g of pure U-235. The specific activity of U-235 is $8.00 \times 10^4 \text{ Bq g}^{-1}$. One gamma-ray line has a branching ratio of 11.0% and an energy of 185.7 keV. A detector is placed 20.0 cm from the pellet. The detector has a circular face with a diameter of 70.0 mm and an intrinsic efficiency of 32.0%. Calculate the expected full-energy peak count rate in counts s^{-1} . Assume isotropic emission and negligible attenuation.

Solution: First, calculate the total activity A of the uranium pellet:

$$A = \text{mass} \times \text{specific activity} = 4.00 \text{ g} \times (8.00 \times 10^4 \text{ Bq g}^{-1}) = 3.20 \times 10^5 \text{ Bq}$$

Next, find the emission rate of the specific 185.7 keV gamma ray by multiplying the total activity by the branching ratio:

$$\text{Gamma Emission Rate} = A \times 0.110 = (3.20 \times 10^5) \times 0.110 = 3.52 \times 10^4 \text{ photons/s}$$

Assuming the detector is a point-like observer viewing a disc of radius $r = 35.0 \text{ mm}$ (3.50 cm) from a perpendicular distance $d = 20.0 \text{ cm}$, the exact geometric efficiency (fractional solid angle) is:

$$\epsilon_{geom} = \frac{\Omega}{4\pi} = \frac{1}{2} (1 - \cos \theta) = \frac{1}{2} \left(1 - \frac{d}{\sqrt{d^2 + r^2}} \right)$$

$$\epsilon_{geom} = \frac{1}{2} \left(1 - \frac{20.0}{\sqrt{20.0^2 + 3.5^2}} \right) = \frac{1}{2} \left(1 - \frac{20}{\sqrt{412.25}} \right) = \frac{1}{2} (1 - 0.985029) \approx 0.0074853$$

The expected full-energy peak count rate C is the product of the gamma emission rate, the geometric efficiency, and the intrinsic efficiency of the detector ($\epsilon_{int} = 0.320$):

$$C = \text{Emission Rate} \times \epsilon_{geom} \times \epsilon_{int}$$

$$C = (3.52 \times 10^4) \times 0.0074853 \times 0.320 = 84.31 \text{ counts s}^{-1}$$

Rounding to appropriate significant figures, the expected rate is $84.3 \text{ counts s}^{-1}$.

4.3 Electromagnetism

Multi-Charge Electric Field

Three charges are placed on the xy -plane at the corners of an isosceles triangle. A $+4 \mu\text{C}$ charge is at the origin. Two $-2 \mu\text{C}$ charges are at $(3, 0, 0)$ and $(-3, 0, 0)$ cm. Find the magnitude of the net electric field E at point P located at $(0, 0, 4)$ cm. Give your answer in MN/C (rounded to the nearest integer). Take $k = 8.99 \times 10^9 \text{ N m}^2\text{C}^{-2}$.

Solution: We convert all coordinates to meters to match standard SI units. Let $q_1 = +4 \times 10^{-6} \text{ C}$ at $(0, 0, 0)$, and $q_2 = q_3 = -2 \times 10^{-6} \text{ C}$ at $(0.03, 0, 0)$ and $(-0.03, 0, 0)$ respectively. Point P is at $(0, 0, 0.04)$. The electric field vector from q_1 points directly outward along the positive z -axis:

$$\vec{E}_1 = k \frac{q_1}{r_1^2} \hat{k} = (8.99 \times 10^9) \frac{4 \times 10^{-6}}{(0.04)^2} \hat{k} = \frac{35960}{0.0016} \hat{k} = 22.475 \times 10^6 \hat{k} \text{ V/m} = 22.475 \hat{k} \text{ MN/C}$$

The electric field from q_2 points from P towards q_2 (due to the negative charge). The vector from q_2 to P is $\vec{r}_{2P} = -0.03\hat{i} + 0.04\hat{k}$. The distance is $r_2 = \sqrt{0.03^2 + 0.04^2} = 0.05 \text{ m}$. The magnitude is:

$$E_2 = k \frac{|q_2|}{r_2^2} = (8.99 \times 10^9) \frac{2 \times 10^{-6}}{(0.05)^2} = \frac{17980}{0.0025} = 7.192 \times 10^6 \text{ V/m} = 7.192 \text{ MN/C}$$

By symmetry, the x -components of $\vec{E}_2$ and $\vec{E}_3$ precisely cancel out, leaving only their combined z -components. The angle θ relative to the z -axis gives $\cos \theta = \frac{0.04}{0.05} = 0.8$. Because the charges are negative on the xy -plane, they pull a positive test charge at P *downward* (negative z -direction):

$$\vec{E}_{2z} = \vec{E}_{3z} = -E_2 \cos \theta \hat{k} = -7.192(0.8)\hat{k} = -5.7536\hat{k} \text{ MN/C}$$

Summing the contributions from q_2 and q_3 :

$$\vec{E}_{23} = 2 \times (-5.7536)\hat{k} = -11.5072\hat{k} \text{ MN/C}$$

The total net electric field is the sum of $\vec{E}_1$ and $\vec{E}_{23}$:

$$\vec{E}_{net} = (22.475 - 11.5072)\hat{k} = 10.9678\hat{k} \text{ MN/C}$$

The magnitude of the net electric field is 10.9678 MN/C. Rounding to the nearest integer yields 11 MN/C.